

Quiz on Compilers

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Question. Consider the following Lex regular expression and propose a transition diagram which allows the recognition of the same lexemes as `num`.

`num` $(\backslash.[0-9]^+|[\+\-]?[0-9]^+(\backslash.[0-9]^+)?)(E[\+\-]?[0-9]^+)?$

Question. Consider the following Lex regular expression and propose a transition diagram which allows the recognition of the same lexemes as `id`.

`id` $[A-Za-z](_*[A-Za-z0-9])^*$

Question. Assume the following input buffer where `␣` represents a blank character.

```
return␣␣x+␣.5E2+y_0
```

Complete the following Lex skeleton

```
%{
#include<string.h>
char* keyword[] = {"else", "if", "return", "then"};
%}
num    (\.[0-9]+| [+\\-]?[0-9]+(\\.[0-9]+)?) (E[+\\-]?[0-9]+)?
id     [A-Za-z]([_]*[A-Za-z0-9])*
```

```
%%
{num}  {

    }
{id}   {
```

```
}
```

```
%%
```

such that the array `keyword` is used¹ and that the output is

```
kwd<return>
id<x>
plus<>
num<.5E2>
plus<>
id<y_0>
```

¹You may use the C function `int strcmp(const char *s1, const char *s2)` which returns 0 if the strings `s1` and `s2` are equal.

Question. Define the meaning of the pointers $\uparrow$, $\downarrow$ and $\updownarrow$ presented in class and show how the input is analysed using the transition diagrams of the previous questions.